TITLE PAGE

- College Name-JJ COLLEGE OF ENGINEERING AND TECHNOLOGY
- Problem Statement –HEALTHTECH
- PS Category- Software/Hardware -SOFTWARE
- Team Name (Registered on portal)-Cure of Thornes
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CONCEPT IMAGE

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HEALTH VAULT

Health Vault – A Portable Digital Lifeline for Migrant Healthcare

💡 Proposed Solution

Health Vault is a secure, multilingual digital health passport that allows migrant workers to carry their complete medical history wherever they go.

It is user-owned, emergency-ready, and accessible anytime, ensuring continuity of care across regions and languages.

🛠 Detailed Explanation

The prototype is a high-performance web application featuring:

🔐 Biometric (Face Lock) secure access

🌐 Multilingual dashboard (English, Malayalam, Tamil)

🚑 One-Tap Emergency View (blood group, allergies, contacts)

🎧 AI Audio Health Briefing for low-literacy users

🤖 AI tools to detect missing records and assist doctors

🎯 How It Addresses the Problem

Prevents loss of medical history during migration

Eliminates language barriers in healthcare

Provides instant emergency access to life-saving data

Ensures continuity of care across hospitals

✨ Innovation & Uniqueness

User-owned digital health identity

AI-powered audio medical summaries

Migrant-specific clinical assistant

Secure, scalable, serverless architecture

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TECHNICAL APPROACH

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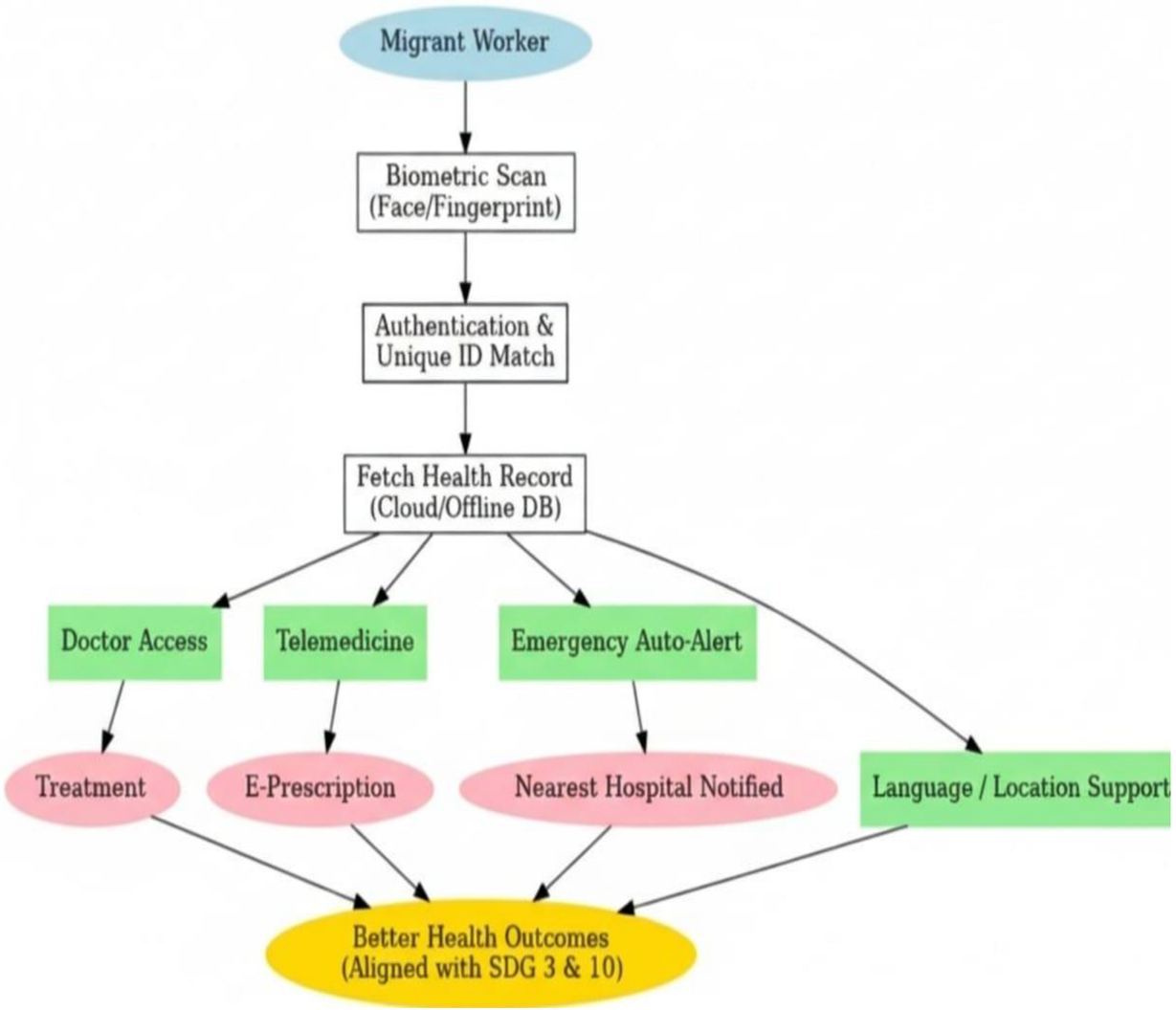
⚙️ **TECHNICAL APPROACH**

📄 **Technologies Used**

TypeScript – Scalable, type-safe development
Next.js 14 & React – High-performance web application
Tailwind CSS & ShadCN UI – Responsive, accessible design
Firebase (Auth + Firestore + Hosting) – Secure serverless backend
Genkit + Google Gemini Models – AI text analysis & audio generation
Zod – Structured medical data validation
PWA Technology – Offline access & mobile compatibility



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FEASIBILITY AND VIABILITY

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✅ Feasibility Analysis

➤**Technical:** Built with modern, scalable technologies (Next.js, Firebase, AI) with PWA offline support.

➤**Operational:** Multilingual, intuitive UI ensures easy adoption. Biometric login and emergency view enhance usability.

➤**Economic:** Serverless hosting reduces costs. Prevents redundant medical tests, saving money for users and hospitals.

⚠️ Potential Challenges & Risks

➤**Data Privacy:** Sensitive medical records must comply with HIPAA/GDPR.

➤**Connectivity:** Low internet access in remote areas.

➤**Trust & Adoption:** Migrants may hesitate to store personal health data digitally.

➤**Institutional Resistance:** Hospitals may resist integrating external patient-owned systems.

➤**AI Reliability:** Accuracy and bias in AI insights.

🛡 Strategies to Overcome Challenges

➤**Security & Compliance**: Biometric authentication, encryption, role-based access.

➤**Offline Capability:** PWA with local caching and automatic sync.

➤**Trust Building:** Emergency view demonstration, NGO partnerships, clear data ownership.

➤**Institutional Collaboration:** API integration and pilot programs.

➤**AI Validation:** Human-in-the-loop and continuous feedback loops.

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IMPACT AND BENEFITS

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POTENTIAL IMPACT

🎯 Potential Impact on Target Audience

✦**MigrantWorkers:**No lost medical historyInstant emergency accessHealth info in their own language

✦**Doctors & Healthcare Providers:**Complete patient recordsAI insights for migrant-specific risksFaster, safer diagnosis

✦**Public Health Systems:**Better disease surveillanceImproved vaccination and chronic disease trackingStronger continuity of care

🌱 **Benefits of the Solution**

Social:

- Promotes healthcare equity
- Reduces marginalization
- Empowers users with health ownership

Economic:

- Reduces redundant tests
- Lowers hospital burden
- Keeps workforce healthy and productive

Environmental:

- Reduces paper usage
- Encourages digital-first sustainable healthcare

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RESEARCH AND REFERENCES

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❖**Esanjeevani telemedicine app:** https://esanjeevani.mohfw.gov.in/#/

❖ABDM app with EHR and ABHA:

https://www.researchgate.net/publication/371091466_The_Role_of_Ayushman_Bharat_Health_Account_ABHA_in_Telehealth_A_New_Frontier_of_Smart_Healthcare_Delivery_in_India

❖Practo and mfine:
https://www.researchgate.net/publication/275110080_Practo_Technologies_the_online_way_of_life